

BLACK-BOX OPTIMISATION

A 12-week search over eight hidden functions

Gaussian Process + UCB, personal-best anchors, and a snap-back rule when the reward falls.

What we are trying to do

Maximise eight unknown functions on the unit box. One query per function per week. No formula and no gradient from the course oracle.

1. Surrogate

Gaussian Process with a Matern kernel and noise. Fit on every (x, y) collected so far.

2. Acquisition

UCB = mean + kappa * std. Kappa starts high once a function looks mature.

3. Action

Draw a local cloud around the personal best. Reject points that are too close to old queries.

4. Update

If y rises, move the centre. If y falls, snap the to the stored best.

Each function has its own model. Dimensions: F1-F2 2D, F3 3D, F4-F5 4D, F6 5D, F7 6D, F8 8D.

From a wide search to a written rule

Weeks 1-2

Wide UCB to look at the cube. Found the F5 high face (last three inputs near 1) and (first input near 0.71).

Weeks 3-4

Tried an SVM region filter and a small neural net hint. They did not change the portal were dropped.

Weeks 5-12

Lock any input that keeps raising y . Move only the free dial. If y falls, the next centre the stored best.

The change came from oracle feedback, not theory. F5 kept scaling on one axis. F2 collapsed when we left the rig

Best observed y after Week 12

Fn	Best y	Week	What actually moves y
1	4.58e-13	2	Still a floor - no usable feature yet
2	0.723	5	Ridge: first input near 0.71, second near 0.01
3	-0.0057	11	Local pocket near 0.55, 0.39, 0.45
4	0.612	11	Tight 4-D pocket around 0.40
5	5214	12	Last three inputs near 1; slide first input up
6	-0.162	7	Fragile; later steps often worse
7	2.486	12	Late local improvement
8	9.996	10	Almost flat near 9.99

Most of the cube is noise. A few inputs carry the variance. Pin the rest and only move the useful dial.

Explore where there is no feature. Exploit where there is one.

What worked

Function 5: lock the high face and only raise the first input. y climbed 4440 to 4520 to 4624 to 4836 to 5214. That is exploitation with one remaining dial.

What failed

Function 2: leaving the second input near 0.24, Week 12 ≈ 0.41) cut y from 0.72 to about 0.24. Extra models in Weeks 3-4 added work, no

Uncertainty rule. When the oracle disagrees with the Gaussian Process, the oracle wins. Do not update the centre of mass. This is how a bad week is absorbed instead of chased.

Balance in practice: higher kappa only on Function 1. Low kappa on Functions 5 and 7. One real evaluation is too expensive to spend on curiosity after a feature is known.

If the search continued

- Keep walking Function 5 on the same face. Do not reopen the last three inputs.
- Restore Function 2 to the Week 5 point (0.712, 0.012) before any new idea.
- Hold Function 7 at the Week 12 best. Treat Function 8 as already flat.
- Leave genuine exploration only on Function 1.

Place in the wider ML landscape

This is Bayesian optimisation for expensive black boxes - the same family used in hyperparameter search. The Gaussian model of the world. UCB is the policy. That split is closer to model-based reinforcement learning than to fitting a large neural network.

For a non-technical stakeholder

We had twelve expensive tests per function. We stopped guessing the whole space once a pattern paid off, and we refused to go back to the start of the search. The clear win is Function 5. The clear warning is Function 2: leaving a working setting alone.